

StrainX - Code Review Summary

Pre-launch. Phase 0 (care tool) is built. Phase 1 (HRV validation study) is built but gated OFF in the database. The system is not live and holds no participant data. This is a code review to finalize IRB wording against exactly what the app collects.

Phase gate - why the study cannot just run

- Study features (video, HRV, Polar H10, self-report, vitals, export) do not exist unless a center is in study_mode.
- Postgres refuses study_mode without ethics_approval_ref + ethics_approved_at (CHECK, migration 0001). Triggers reject every study write otherwise (0003, 0007).
- Review before approval: a POC placeholder ref via scripts/set-study-mode.mjs --poc (never real data). Proof: supabase/tests/phase_gate_smoke.sql.

Study data map - what the IRB must cover

Data element	Captured on	Stored in	Read by	Retention
Raw finger/face video	kiosk	ppg-video bucket (private)	staff only	whole study
PPG waveform + timestamps	kiosk	ppg-raw bucket (private)	staff only	whole study
Time-domain HRV (HR, RMSSD, SDNN)	kiosk (on-device)	study_recordings	staff	whole study
0-10 stress self-report (after)	kiosk	study_self_reports	staff	whole study
Vitals (BP, temp, weight, sleep, activity)	staff / kiosk	study_recording_vitals	staff	whole study
Covariates (age, sex, height)	staff	study_participant_profile	staff	whole study
Polar H10 reference RR stream	H10 (BLE or offline)	ppg-raw + ref_start_marker	staff	whole study

De-identified export uses coded ids (P001, ...), never names. Audio is never captured (video is recorded with audio off; there is no audio bucket). emergency_contacts is never touched by any automated path. The 0-10 self-report is a research anchor only: never shown on a care screen and never fed into the care signal (strain_index).

Where to review

- Gate + schema: supabase/migrations/0001, 0007; supabase/tests/phase_gate_smoke.sql
- Access control (RLS): 0006 + 0007 - the kiosk is write-only (no SELECT on any capture table)
- Study capture: apps/kiosk/src/screens/StudyCaptureScreen.tsx; lib/ppg.ts; lib/h10.ts
- Staff review / covariate entry / export: apps/web/src/app/staff/records/page.tsx
- Analysis export: scripts/export-fantasia.mjs | Care-language guard: scripts/copy-lint.mjs

Build status (honest)

- Built + checked here (lint:copy, web tsc): schema + gate + RLS + both private buckets, kiosk study capture (video + waveform + HRV + H10 + self-report + finger/face/light/session), staff review/entry/export, POC opener, Fantasia export.
- Needs a real environment: apply 0007 to a live Postgres + run the smoke test; a kiosk device build (native camera/BLE); the on-device video upload uses fetch(file://) and may need expo-file-system.
- Inherently external: Fantasia field-name reconciliation, the H10 off-load format, session-length tuning.

Not live: study_mode ships false; no real participant data. Full detail in REVIEW.md. Prepared 2026-08-08.